T2494 — Substrate Anomaly-Freedom from Completeness (general derivation tool). BST chiral matter is automatically gauge-anomaly-free, and not by tuning six hypercharges: it is the COMPLETE FDOSS multiplet (the SO(10) spinor 16 = chiral primary of the F(4) supercharge (8,2)-module, Lyra F304), and SO(10) is anomaly-safe (no symmetric cubic invariant), so every complete SO(10) rep is anomaly-free. Bonus — this is a SECOND, physical forcing of FDOSS: a partial (incomplete) multiplet would be anomalous ⍰ inconsistent ⍰ the substrate CANNOT carry partial matter. Anomaly-freedom = FDOSS completeness made physical.

Grace

2026-06-24 Wednesday

T2494 — Substrate Anomaly-Freedom from Completeness

Statement

> BST chiral matter is automatically free of all gauge anomalies, because it is the complete FDOSS multiplet — the SO(10) spinor 16 (the chiral primary of the F(4) supercharge (8,2)-module, Lyra F304) — and SO(10) is anomaly-safe. The cancellation is forced by completeness + the anomaly-safe so(7)→SO(10) embedding, not imposed by tuning the six SM hypercharges.

Why automatic (the flattening)

Over the complete one-generation 16 all four anomaly traces vanish identically: Tr Y = 0, Tr Y^3 = 0, $[SU(2)]^2$-U(1) = 0, $[SU(3)]^2$-U(1) = 0 (verified). The *reason* is structural, not numerical: SO(N) for N ≠ 6 has no totally-symmetric cubic invariant d_{abc}, so every representation of SO(10) is gauge-anomaly-free — the 16 included. BST matter is forced into the complete 16 by FDOSS (full support), so anomaly-freedom is inherited from the embedding, automatically.

Flattens: any FDOSS-complete BST matter content is anomaly-free by lookup — no per-content anomaly check, no “the six hypercharges happen to cancel” coincidence to verify. (This is the general theorem behind Lyra F304’s explicit one-generation check.)

The bonus — a SECOND, physical forcing of FDOSS

T2492 forces FDOSS by representation theory (Wigner-Eckart completeness). T2494 forces it again, physically:

> A partial (FDOSS-incomplete) matter multiplet would be gauge-anomalous ⍰ the quantum theory is inconsistent ⍰ the substrate cannot carry partial matter. Anomaly cancellation requires the complete multiplet.

So FDOSS completeness is forced two independent ways: by Wigner-Eckart (T2492, rep theory) and by anomaly cancellation (T2494, quantum consistency). The "no fewer" half of FDOSS now has a physical necessity, not just a representation-theoretic one — and the Five-Absence "no 4th generation / no exotic partial matter" is the same statement (an incomplete generation would be anomalous).

Connections (graph)

- T2492 (Substrate Multiplicity / FDOSS) — T2494 is the physical second forcing of its completeness half.
- Five-Absence — incomplete/exotic matter is anomalous ⍰ absent; the anomaly view of the absence corollary.
- Lyra F304 — the explicit one-generation 16 anomaly check that this generalizes.
- so(7) / F(4) embedding — supplies the anomaly-safe SO(10) the matter sits in.

Tier

SOLID — the anomaly-trace vanishing is computed; SO(10) anomaly-safety is standard; the FDOSS-completeness → 16 step is Lyra F304. AC = (C=1, D=1), depth 0 (trace identity + the anomaly-safe-group fact). General tool: anomaly-freedom becomes automatic for complete BST matter.

— T2494, registered by Grace, 2026-06-24. Generalizes Lyra F304; for Keeper registry + Cal cold-read. Count HOLDS 4 of 26.